Lam Fung Academy · LF Academy

Primary 5 · Semester 2 · 24th Lesson · Student Handout

# Fraction word problem special project

Unit 3 · Remaining problem ＋ Inverse towardwards reasoning · 65 minutes · 1-to-3 Online Lesson

**Corresponding textbooks:**"New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 Comprehensive + Modern Education 5L B Unit 13

**Core traps:**□ T3 misjudgment of fractional division word problems · T7 "What fraction is left" is mistaken for "fraction of the original number"

**SSPA Related:** 🔴 Extremely High Frequency Paper 1 accounts for about 20%, Paper 2 accounts for about 15%

**Prerequisite knowledge:**Chapter 7-8 (addition and subtraction) · Chapter 9 (multiplication) · Chapter 22 (division) · Chapter 23 (mix of four)

**The goal of this class:**❶ Distinguish the operation types of fraction word problems ❷ Identify the "remaining fraction" trap ❸ Multi-step reasoning ❹ Reverse reasoning (reverse the original number from the result)

Student Name: ____________ Class: ____________ Date: ____________ Time Spent: ____________

## I.Warm-Up Questions ( total 4 question · 5 minutes )

| # | Question | Difficulty | Working Space ( Show full working ) |
|---|---|---|---|
| 1 | Convert "A bottle of juice has $\frac{2}{3}$ liters, drink $\frac{1}{4}$ liters" into a formula. (Just list the formula, no calculation required) | Basic | |
| 2 | "Divide $\frac{3}{4}$ kilograms of candy equally among 5 people" - what operation is this? Column formula. | Basic | |
| 3 | "Used full-length \|\|\|SEP\|\|\|" - What fraction of the full length is the rest? (Tip: full length = 1) $\frac{1}{3}$ "——What fraction of the total length is the rest? (Tip: full length = 1) | Basic | |
| 4 | Are "remaining \|\|\|SEP\|\|\|" and "original \|\|\|SEP\|\|\|" necessarily the same? Give examples. $\frac{1}{2}$ " and “Original $\frac{1}{2}$ "Is it necessarily the same? Give examples. | Advanced | |

## II.Core Knowledge + Worked Examples

### Knowledge Point 1: One-Step Fraction Problems—Identify Operation Types 🔴 SSPA

**Keywords → Operation comparison table:**

① **"total" "total" "total" → addition**(total different parts)

② **"remainder" "remaining" "difference" → subtraction**(deduct a part from the total)

③ **(Find the fraction of a quantity)**"Equal share" "Each portion" "How much can be..." → Division

④ **(distribute equally or include division)**(divide equally or include division)

| + | − | × | ÷ |
|---|---|---|---|
| **addition**<br>total, total combined, total | **Subtraction**<br>Remaining, remaining difference, used | **multiplication**<br>Several times or parts of... | **division**<br>How much can be divided equally, each... |

**Example 1 - Addition Identification**

**Xiaomei has $\frac{1}{2}$ boxes of chocolates, and her mother buys $\frac{2}{3}$ boxes. How many boxes of chocolate does she have in total?**

**Example 2 - Subtraction Identification**

A bottle of juice originally contains $\frac{5}{6}$ liters, and drank $\frac{1}{3}$ liters. How many liters are left?

### Knowledge Point 1 Synchronized practice (circle key words → column → calculate → answer sentence)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| 5 | The cake shop sold $\frac{2}{5}$ cakes in the morning and $\frac{1}{3}$ cakes in the afternoon. How many cakes were sold throughout the day? | 🌱 | |
| 6 | The original water in the water bottle is $\frac{3}{4}$ liters, pour out $\frac{1}{5}$ liters. How many liters are left? | 🌱 | |
| 7 | A rope is $\frac{2}{3}$ meters long, and the full length of $\frac{1}{2}$ is used. How many meters were used? (Note: It means "using ½ of the full length", not "using ½ meter"!) | 🌿 | |
| 8 | $\frac{3}{5}$ Kilogram of sugar, divided equally into 3 small bags. How many kilograms of sugar are in each bag? | 🌿 | |

⚠ **Q7 demonstrates the key trap: "Using ½ of the full length" vs "Using ½ meter" - the former is multiplication (finding the part), the latter is subtraction (deducting the length). If the judgment is wrong, the entire question will be wrong!**

### Knowledge point 2: "The remaining fraction" trap 🔴 SSPA extremely high frequency killer question

**This is the most common question type to get wrong in the full test!**

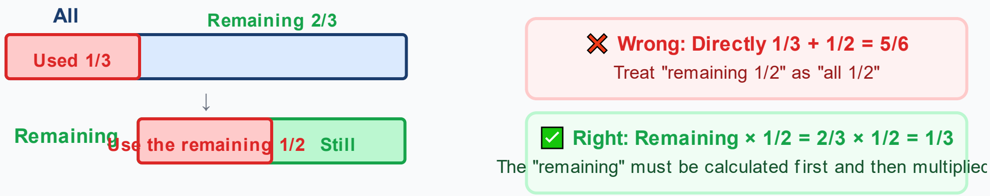
① **"Remaining a/b" ≠ "Original a/b"**- You must first find the remaining amount, and then take its fraction

② Solution: First find**Remaining**= Original − Used/Eat/Spent

③ Find again**The remaining fraction**= Remaining × Fraction

④ **Never multiply directly by the original number!**

"The remaining fraction" trap diagram




**Trap detonation example 3 (the most important demonstration in this class!)**

Aribbon is $\frac{3}{4}$ meters long. After using $\frac{1}{3}$ meters, use the remaining $\frac{1}{2}$ to wrap gifts. How much rice did it take to wrap the gift?

**❌ Fatal errors (80% of students)**

$$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8} \text{rice}$$

Just calculate the "remaining ½" as "½ of the original ribbon"!

**✅ Correct solution**

$$\text{remaining} = \frac{3}{4} - \frac{1}{3} = \frac{5}{12} \rightarrow$$

$$\text{Wrapping gifts} = \frac{5}{12} \times \frac{1}{2} = \frac{5}{24} \text{ rice}$$

You must first find the remainder, and then take the remaining ½ (not ½ of the original ribbon!)

**Tip: "When you see "remainder", first subtract and then multiply - first subtract the remainder and then multiply the fraction. Circle the three words "remainder" and be careful not to multiply the original number directly!"**

**Example 4 - Multiplicative "What fraction is left"**

Xiao Ming has pocket money |||SEP|||. After spending |||SEP|||, he saves the remaining |||SEP|||. How much have you saved? (Answer with mixed fractions) $\frac{3}{5}$ ,it took $\frac{1}{5}$ Finally, put the remaining $\frac{2}{3}$ Savings. How much have you saved? (Answer with mixed fractions)

## Knowledge Point 2 Synchronous practice (circle "the remaining" → first find the remaining → again take fraction → answer sentence)

| # | Question | Difficulty | Working Space |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **9** | A bottle of orange juice $\frac{7}{8}$ liters, drank $\frac{1}{4}$ liters. The remaining $\frac{1}{3}$ is used to make jelly. How many liters are used to make jelly? | 🌿 | |
| **10** | A bag of sugar weighs $\frac{4}{5}$ kilograms. After using $\frac{1}{5}$ kilograms, the remaining $\frac{1}{2}$ is used to make cakes. How many kilograms were used to make the cake? | 🌿 | |
| **11** | A rope is $\frac{5}{6}$ meters long. Cut off $\frac{1}{3}$ meters first, and use the remaining $\frac{2}{5}$ to make knots. How many meters did it take to make the knot? | 🌳 | |
| **12** | A box of juice $\frac{3}{4}$ liters, drank $\frac{1}{2}$ liters. Then divide the remainder equally among 3 friends. How many liters does each person receive? | 🌳 | |
| **13** | The sister had a box of chocolate \|\|\|SEP\|\|\|. After eating the $\frac{5}{8}$ box, the remaining $\frac{1}{4}$ was given to her brother. How many boxes does the brother get? (Note: There are two traps in the question - subtract first and then multiply!) $\frac{3}{4}$ Gave it to my brother. How many boxes does the brother get? (Note: There are two traps in the question - subtract first and then multiply!) | 🌳 | |

⚠️ **The second biggest trap: "divide the remainder equally" = first subtract (find the remainder), then divide (divide equally). Not divide first and then subtract! In the wrong order, the answer is completely different.**

### Knowledge point 3: Multi-step reasoning text questions 🔴 SSPA advanced must test

**A question contains two or more operational steps, which need to be executed in the correct order:**

① **Circle all key data**(quantity, fraction, relational words)

② **Draw process arrow |||SEP|||: original → first step (+−×÷) → intermediate result → second step (+−×÷) → answer**|||SEP||| (Add parentheses when necessary)

③ **Calculate**step by step, check whether it is reasonable at each step (for example, the remaining value cannot be greater than the original)

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### Example 5 – Addition, Subtraction and Mixing

**The original water in the bucket is $\frac{5}{6}$ liters. After using $\frac{1}{3}$ liters, add $\frac{1}{2}$ liters. How many liters of water are there now? (need to subtract first and then add)**

**Example 6 - Mixed multiplication and division**

A rope is $\frac{2}{3}$ meters long. After using the full length of |||SEP|||, divide the remaining length into 3 equal parts. How many meters long is each section? $\frac{1}{4}$ Finally, divide the remaining pieces into 3 equal parts. How many meters long is each section?

### Knowledge Point 3 Synchronous practice (draw process → column → step by step calculation → answer sentence)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| 14 | A bottle of juice $\frac{3}{4}$ liters, drank $\frac{1}{4}$ liters in the morning, and added $\frac{1}{2}$ liters in the afternoon. How many liters are there in total now? | 🌿 | |
| 15 | A box of cakes weighs $\frac{5}{8}$ kg. After cutting off $\frac{1}{4}$ kilograms, divide the remainder equally among 3 people. How many kilograms does each person get? | 🌳 | |
| 16 | Xiaohua has a $\frac{3}{5}$ premium, and Xiao Ming has $\frac{1}{3}$ 2 times the premium. How many liters of water do they have in total? | 🌳 | |
| 17 | A rope is $1\frac{1}{2}$ meter long. After using the full length of \|\|\|SEP\|\|\|, use the remaining $\frac{2}{5}$ for decoration. How much rice was used to make the decoration? $\frac{1}{3}$ Used for decoration. How much rice was used to make the decoration? | 🌳 | |

## Knowledge Point 4: Reverse Reasoning Word Questions 🔴 SSPA Killer Questions

**Infer the original quantity from the result - present the score question of Paper 2!**

① **Identification |||SEP|||: The question gives the "final result" and asks you to find "how many there are"** Strategy |||SEP|||: From the final result

② **Reverse operation** (addition, subtraction, subtraction, addition, multiplication, transformation, division, division, transformation and multiplication) **Formula**: Forward sequence formula, solve the problem in reverse direction - reverse each step in the forward direction

③ **Check calculation**: Use the calculated original number, calculate it once in the forward direction, it should be equal to the result given in the question

④ **Check calculation**: Use the calculated original number and calculate it once, it should be equal to the result given in the question.

| → | ← | ✓ | ⚠ trap |
|---|---|---|---|

| Thinking in the direction | reverse operation | Check calculation | |
|---|---|---|---|
| original → subtraction → remainder → × fraction | Infer from the result that ÷ becomes ×, + becomes − | Use the original number to calculate and confirm | Pay attention to unit and fraction types |

**Example 7 - Reverse Reasoning (Subtraction Reverse)**

**Xiao Ming had some chocolates. After eating $\frac{1}{4}$ boxes, there were $\frac{1}{2}$ boxes left. How many boxes of chocolate were there?**

**Example 8 - Reverse Reasoning (Hybrid Reverse)**

**From a rope, first cut off $\frac{1}{3}$ meters, and then use the remaining $\frac{1}{2}$ for handicrafts. For handicrafts, $\frac{1}{4}$ meters are used. How many meters long was the rope?**

### Knowledge Point Four Synchronization Practice

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| 18 | A bottle of juice, after drinking $\frac{1}{3}$ liters, there will be $\frac{1}{4}$ liters left. How many liters did it originally have? | 🌿 | |
| 19 | Some candies are divided equally among 4 people, and each person gets $\frac{1}{6}$ kilograms. How many kilograms of candy were there? | 🌿 | |
| 20 | A box of cakes, eaten $\frac{1}{2}$ boxes, and then eaten $\frac{1}{3}$ boxes, leaving $\frac{1}{6}$ boxes. How many boxes of cakes were there? | 🌳 | |

**🧠 Tip (reverse reasoning): "Calculate backwards from end to beginning, add, subtract, multiply, and divide, subtract, add, divide, multiply, and remember to check it after you finish the calculation."**

## III. Lesson Layered Synchronization Practice

### Basic layer (total 5 questions, all must do, all text questions)

| # | Question | Difficulty | Working Space |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **21** | Xiaomei has $\frac{2}{3}$ meters of ribbon, and Xiaohua has $\frac{1}{4}$ meters. How many meters of ribbon do they have in common? | 🌱 | |
| **22** | A bottle of water has $\frac{3}{5}$ liters and uses $\frac{1}{10}$ liters. How many liters are left? | 🌱 | |
| **23** | The $\frac{1}{3}$ of the cake is chocolate flavor, and the rest is strawberry flavor. What percentage of the cake is strawberry flavor? | 🌱 | |
| **24** | $\frac{3}{4}$ Kilograms of flour, make one cake per $\frac{1}{8}$ kg. How many cakes can be made? | 🌱 | |
| **25** | A rope is $\frac{4}{5}$ meters long. After using $\frac{1}{5}$ meters, the remaining $\frac{1}{2}$ How many meters have been left? | 🌿 | |

### Advanced layer (total 4 questions, 🧗🚀 choose do)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| **26** | A bottle of juice $\frac{5}{6}$ liters. After pouring $\frac{1}{2}$ liters, divide the remaining juice equally among 4 people. How many liters does each person get? | 🌿 | |
| **27** | A bag of rice weighs $\frac{7}{8}$ kg. $\frac{1}{4}$ kilograms were used to cook rice, and the remaining $\frac{2}{3}$ were used to cook porridge. How many kilograms were used to cook the porridge? | 🌿 | |
| **28** | My brother used $\frac{1}{3}$ of his pocket money to buy books, and then used the remaining $\frac{1}{2}$ to buy stationery. He spent $\frac{10}{1}$ yuan to buy stationery. How many yuan was it originally? | 🌳 | |
| **29** | There are $\frac{3}{5}$ liters of juice in the bottle. After pouring $\frac{1}{4}$ liters of water, pour out $\frac{1}{3}$ of the mixture. How many liters were poured out? | 🌳 | |

### 🌳 challenge layer (total 2 questions, 🚀 choose do, SSPAKiller Questions)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| **30** | For a rope, first cut off \|\|\|SEP\|\|\|, then cut off the remaining \|\|\|SEP\|\|\|, and finally there will be 3 meters left. How many meters long was the rope? (First find what fraction of the original amount is left after the second cut, and then work in reverse) $\frac{1}{3}$, and then cut off the rest $\frac{1}{2}$, and finally 3 meters remain. How | ⛰️ | |

| | | | |
|---|---|---|---|
| | many meters long was the rope? (First find what fraction of the original amount is left after the second cut, and then work in reverse) | | |
| 31 | Xiao Ming spent $\frac{2}{5}$ of his pocket money to buy snacks, and then spent the remaining $\frac{1}{3}$ to buy stationery, and finally he was left with $\frac{4}{1}$ yuan. How much is the original pocket money? | ⛰ | |

## IV.⛰ The ultimate challenge — SSPAPaper 2 practical question

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| ⛰ 1 | \|\|\|SEP\|\|\| on the bookshelf is for Chinese books, $\frac{1}{4}$ is for English books, and the remaining $\frac{1}{3}$ is for math books. If mathematics books occupy \|\|\|SEP\|\|\|, what fraction of the total space on the bookshelf is occupied by books? (Check: Chinese + English + Mathematics = total number of books) $\frac{1}{2}$ Put the math book. If math books took up bookshelf $\frac{5}{24}$, what fraction of the total space on the bookshelf is occupied by books? (Check: Chinese + English + Mathematics = total number of books) | ⛰ | |
| ⛰ 2 | A barrel of oil contains $\frac{5}{6}$ liters. The first time I used $\frac{1}{4}$ liters, the second time I used the remaining \|\|\|SEP\|\|\|, and the third time I used the remaining $\frac{2}{3}$ of the second time. How many liters are left at the end? $\frac{1}{2}$. How many liters are left at the end? | ⛰ | |

### Comprehensive reasoning question (🚀Choose do, Paper 2 points question)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| R1 | For a bottle of water, I used $\frac{1}{4}$ on the first day, used the remaining $\frac{1}{3}$ on the second day, and ended up with $\frac{1}{2}$ liters left. How many liters of water was there? (Hint: The remaining ⅓ of the first day was used on the second day, that is, ⅓ of the "original ¾" was used on the second day = the original ¼) | ⛰ | |
| R2 | A, B, and C share candies. A ate \|\|\|SEP\|\|\|, B ate the remaining \|\|\|SEP\|\|\|, and C ate the last remaining box (\|\|\|SEP\|\|\| box). How many boxes of candy were | ⛰ | |

| | | | |
|---|---|---|---|
| | there? $\frac{1}{3}$, B ate the rest $\frac{1}{2}$, B ate all that was left ($\frac{1}{4}$ box). How many boxes of candy were there? | | |

## V. Class afterhomework

### Basic must-do questions (total 6 questions, all text questions, must write answer sentences)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| H1 | A bottle of orange juice $\frac{3}{4}$ liters, my brother drank $\frac{1}{4}$ liters, and my sister drank $\frac{1}{3}$ liters. How many liters did they drink in total? | 🌱 | |
| H2 | A rope is $\frac{4}{5}$ meters long, cut off $\frac{1}{10}$ meters. How many meters are left? | 🌱 | |
| H3 | $\frac{2}{3}$ Kg of sweets, one pack per $\frac{1}{6}$ kilogram. How many bags can be packed? | 🌱 | |
| H4 | A pancake, the sister ate $\frac{1}{3}$ and the brother ate \|\|\|SEP\|\|\|. How much of the pancake is left? $\frac{1}{4}$. How much of the pancake is left? | 🌱 | |
| H5 | A bag of sugar weighs $\frac{5}{8}$ kilograms. After using $\frac{1}{4}$ kilograms, the remaining $\frac{1}{2}$ is used to make desserts. How many kilograms were used to make the dessert? | 🌿 | |
| H6 | A ribbon is $\frac{3}{5}$ meters long, and the full length of $\frac{1}{3}$ is used. How many meters were used? | 🌿 | |

### For advanced, choose doquestion (total 2 questions, 🚀 choose do)

| # | Question | Difficulty | Working Space |
|---|---|---|---|
| H7 | A rope is $1\frac{1}{2}$ meter long. After using $\frac{1}{2}$ meters, the remaining $\frac{1}{4}$ is used to make knots. How many meters did it take to make the knot? | 🌳 | |
| H8 | Xiao Ming had pocket money, so he spent $\frac{1}{3}$ to buy stationery, and then spent the remaining $\frac{1}{2}$ to buy snacks, and finally he was left with \|\|\|SEP\|\|\|. How much is the original pocket money? (Expressed as a fraction, assuming original pocket money = 1) $\frac{1}{3}$. How much is the original pocket money? (Expressed as a fraction, assuming original pocket money = 1) | ⛰️ | |

## VI. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

- ☐ I know the pitfalls of solving each knowledge point
- ☐ I can complete 🌱basic questions independently
- ☐ I can challenge 🌿advanced questions
- ☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

- ☐ Identify all trap types in our hall
- ☐ Solve 🌱basic questions independently (100% correct)
- ☐ Challenge 🌿Advanced questions (80%+ correct)
- ☐ Explain the lesson formula to classmates

| # | common error | Correct Approach |
|---|---|---|
| 1 | **"Used ½" vs "Used ½ meter" confusion \|\|\|SEP\|\|\|: The former is a multiple (multiplication), the latter is a quantity (subtraction)**: The former is a multiple (multiplication), the latter is a quantity (subtraction) | See clearly w hether it is "a fraction of the total length" or "how many meters". "" is mostly multiplication. |
| 2 | **"Remaining a/b" is calculated as "original a/b"** | You must first find the remaining quantity, and then take the remaining a/b! This is the most important lesson in the w hole class. |
| 3 | **Direct addition and subtraction of fractions of the remaining amount \|\|\|SEP\|\|\|: The remaining ½ and the remaining ⅓ are directly added**: The remaining ½ and the remaining ⅓ are added directly | The "remainder" is based on a different basis each time, and must be calculated individually and then added together. |
| 4 | **The order of multi-step questions is wrong**(Add first and then multiply as multiplication first and then add) | Circle keyw ords and draw process arrow s to confirm the calculation order. |
| 5 | **Inverse questions will not be "operated backwards"** | Forw ard continuation, solve the problem in reverse: + changes to −, − changes to +, × changes to ÷, ÷ changes to × |
| 6 | **Missing sentences/missing units** | Text questions must w rite "Answ er:..." + complete unit, otherw ise step points w ill be deducted |
| 7 | **The fraction result is unreduced** | The final step must check reduction (HCF) and improper fractions → mixed numbers |